ENRIQUE GARCIA RIVERA

e.garciarivera@wustl.edu | linkedin.com/in/enrique-gr | github.com/enrique-gr | enrique-gr.github.io/portfolio

EDUCATION

Washington University in St. Louis, McKelvey School of Engineering *Expected: May 2027*
B.S. in Mechanical Engineering, Minors in Aerospace Engineering, Robotics, and Mechatronics GPA: 3.64/4.00
Relevant Coursework: Control Systems (MATLAB/Simulink), Fluid Mechanics, Heat Transfer, Thermodynamics, Solid Mechanics, Vibrations, Design of Thermal Systems, Numerical Methods & Matrix Algebra (MATLAB), Materials Science

SKILLS

- **Engineering Competencies:** Mechanical design, structural FEA, rapid prototyping (3D printing), precision alignment, load path analysis and verification, test planning and execution, experimental data analysis.
- **Software & Tools:** SolidWorks, ANSYS Mechanical, MATLAB/Simulink, Python, C++, embedded systems (Teensy/Arduino), sensor integration (IMU, GPS), electronic bench testing (oscilloscope, multimeter).

EXPERIENCE

WashU VTOL – *Avionics & Systems Integration Lead*, St. Louis, MO Sep 2025 – Present
- Founding member of WashU's inaugural VTOL team, defining avionics architecture and electrical topology for the Vertical Flight Society Design-Build-Vertical Flight Competition.
- Designed and integrated a Pixhawk 6C flight controller with a Raspberry Pi companion computer to enable autonomous mission execution, geofencing, and onboard mission logic.
- Designed separated propulsion and avionics power systems with external kill-switch compliance; conducted bench validation of telemetry, RC links, and failsafe behavior.
- Collaborated with mechanical and software teams to optimize component placement, vibration isolation, thermal management, and PX4 parameter tuning; validated system behavior through HIL simulation and ground testing.

Robotics Lab, UMKC – *Research Intern (Grant-Funded)*, Kansas City, MO May 2025 – Aug 2025
- Developed a real-time embedded sensing platform (Teensy 4.0) synchronizing 1 kHz IMU and EMG data streams for closed-loop robotic applications.
- Characterized sensor noise, bias drift, and timing jitter to establish hardware performance bounds and inform state-estimation design.
- Implemented an Extended Kalman Filter in MATLAB for multi-sensor fusion, reducing state-estimation drift during dynamic motion.
- Built Python-based validation tools to quantify estimation error and system repeatability; research published at IEEE Body Sensor Networks (BSN) 2025.

WashU Design/Build/Fly – *Aerodynamics & Payload Engineer*, St. Louis, MO Sep 2024 – Present
- Designed wing geometries and control surfaces using XFLR5 and analytical methods, balancing aerodynamic performance with manufacturability constraints.
- Performed structural FEA in ANSYS to verify wing spars met 2.5g load requirements; correlated simulation results with physical load tests.
- Prototyped payload release mechanisms through iterative CAD and 3D printing, identifying and eliminating mechanical failure modes before flight testing.
- Documented analysis procedures to maintain design continuity and enable knowledge transfer across team cycles.

Washington University McKelvey School of Engineering – *Course Assistant*, St. Louis, MO Aug 2025 – Dec 2025
- Supported students in MEMS 2000: Electrical & Electronic Circuits for Mechanical Engineers through lab sessions, weekly office hours, and troubleshooting circuit designs, oscilloscope measurements, and breadboard implementations.
- Graded lab reports and assignments, providing detailed feedback on experimental methodology and the quality of technical documentation.

FedEx Corporation – *Material Handler*, Kansas City, MO Jun 2024 – Aug 2024
- Maintained a throughput of over 100 packages per hour while following rigorous safety protocols in a high-tempo logistics environment.
- Identified loading inefficiencies through team communication, implementing improvements reducing handling errors.

TECHNICAL PROJECTS

Sentinel - Kinetic Intercept Prototype (C++ / Python): Independently studied computer vision and control theory to build a target-tracking robot. Implemented a Kalman Filter on a microcontroller to fuse sensor data and direct a 2-axis gimbal.

Hyperion - Orbital Physics Simulator (C++): Developed a 6-DOF physics engine from scratch using the J2 Perturbation Model to explore orbital mechanics and numerical integration methods.

Flight Nav EKF System (Embedded C++): Explored sensor fusion by writing a 12-state Extended Kalman Filter that combines GPS, barometer, and IMU data on resource-constrained hardware.